

The Efficient Frontier of Dollar-Cost Averaging

One-page summary. Jaggernaut Wealth Services, June 2026.

The setup

A hypothetical investor at the 75th percentile of full-time income saves 20% of gross pay monthly and dollar-cost-averages it into one portfolio over 1980–2026 (46 years). 56 strategies — static allocations, age-based *glides* (designs that shift the allocation over time — e.g., from stocks toward bonds, gold, and/or cash as retirement approaches), and momentum overlays — plus a wind-down-to-cash strategy are each run in a fully taxable and fully tax-free account, on the actual history and on 1,000 block-bootstrap simulations. Strategies are ranked on the distribution of outcomes, not the single realized path.

The two axes

- **Median wealth** — the middle of the 1,000 simulated final balances.
- **Terminal pain** — the 10th-percentile drawdown in the final five years before retirement. Capped at –30%: deeper late-life loss is judged unrecoverable.

A strategy is **efficient** if no other beats it on both wealth and pain within the –30% cap.

Key findings

- Median final balances ran **\$2.0M to \$8.7M** across strategies and accounts.
- High-wealth mixes (\$5.7–8.7M) carried the deepest drawdowns (–37.8% to –59.1%). No strategy had both the most wealth and the smallest drawdown.
- In the worst joint stock/bond years (2008, 2002, 2022), a diversified stocks/gold/bonds mix fell far less than 60-Stk/40-Bnd.
- Widening the record to 1962–2026 and measuring in real 2026 dollars: bond-heavy strategies fare worse in taxable accounts on inflation-adjusted wealth. A stocks/gold/bonds glide ends with **about 24% more real wealth** than the bond-heavy target-date glide in taxable; tax-free is a tie.
- Taxes lowered wealth by **15–30%** per strategy vs. the tax-free account. Hard rebalancing realizes gains, so it ranks high in tax-free and lower in taxable.
- Efficient strategies within the –30% cap: **six in taxable, five in tax-free** (the tax-free set includes all three wind-down variants).

The efficient set

Taxable account

Strategy	Median	Pain
TDF 90Stk/10Bnd→30Stk/70Bnd (hard)	\$3.04M	–19.3%
TDF 90Stk/10Bnd→30Stk/30Gld/40Bnd (hard)	\$3.04M	–22%
All Weather, no rebal: 30Stk/40LongT/15Bnd/15Gld	\$3.81M	–29%
All Weather, soft: 30Stk/40LongT/15Bnd/15Gld	\$3.32M	–25%
Glide D, soft: 50/25/25 → 40/30/30 → 30/30/40 (Stk/Gld/Bnd)	\$3.46M	–28%
Permanent Portfolio, hard: 25Stk/25Gld/25LongT/25Cash	\$2.02M	–18.3%

Tax-free account (Roth/401k)

Strategy	Median	Pain
Permanent Portfolio, hard: 25Stk/25Gld/25LongT/25Cash	\$3.00M	–16%
TDF 90Stk/10Bnd→30Stk/70Bnd (hard)	\$4.93M	–18%
Wind-down to cash, sell 60% five years out (then buy stock)	\$7.51M	–24%
Wind-down to cash, sell 50% five years out (then buy stock)	\$7.77M	–25%
Wind-down to cash, sell 40% five years out (then buy cash)	\$7.90M	–27%

TDF 90/10→30/70 (hard) and the Permanent Portfolio (hard) appear in both sets. Sets unchanged under a second random seed.

What the report concludes

Tax-free: the wind-down-to-cash strategies have the highest wealth-per-unit-pain — ride 100% stocks through the accumulation, then sell to cash in the five years before retirement. All three sizing variants (sell 40%, 50%, or 60% five years out) end richer than any glide and stay within the pain cap; sell-60% has the top W/TP. Taxable: the hard-rebalanced 90/10→30/70 target-date glide leads on wealth-per-pain, but a diversified stocks/gold/bonds glide (Glide D) delivers more real wealth once inflation is measured over a longer historical record. High-equity glide (target-date) requires the hard variant; the soft version overshoots the cap. The Permanent Portfolio remains the calmest efficient choice in both accounts.

Stk = US stocks; Bnd = intermediate bonds; LongT = long Treasuries; Gld = gold; Cash = short Treasuries. “Hard” rebalances yearly (realizing taxable gains); “soft” redirects new contributions only.

Source: The Efficient Frontier of Dollar-Cost Averaging, Jaggernaut Wealth Services, LLC, June 2026. Full report at https://github.com/jsrini/DCA_EfficientFrontier/blob/main/wealth_report.pdf; data, code, and build pipeline in the same repo. Backtest and thought experiment, not financial advice. Figures nominal 2026 dollars unless labeled real; AI-written code, spot-checked — reproduce before relying.